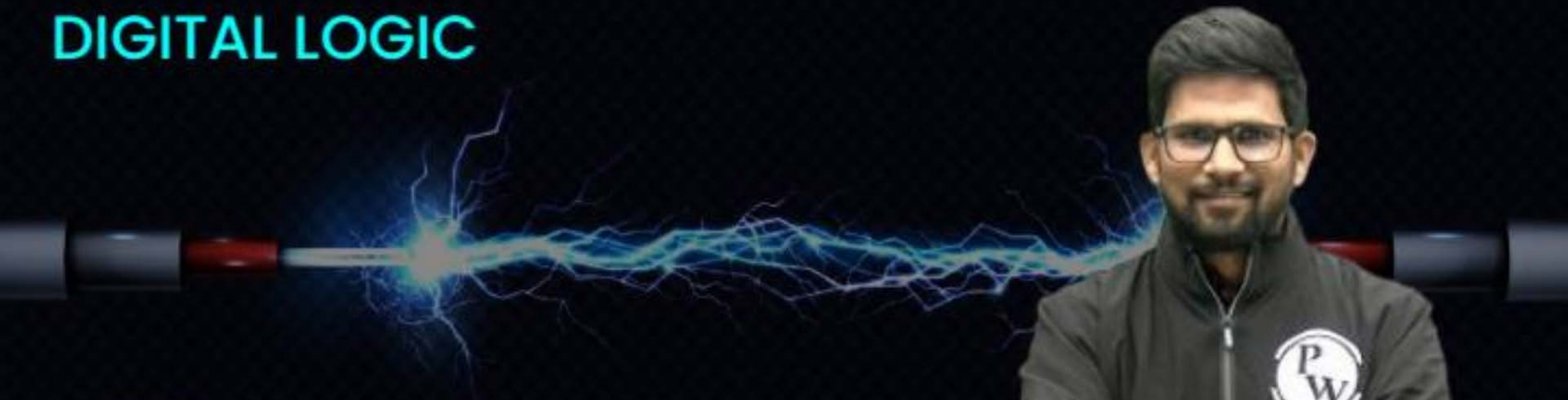
COMPUTER SCIENCE & IT

DIGITAL LOGIC





Lecture No: 08

Miscellaneous Topics

By- Chandan Gupta Sir

Recap of Previous Lecture



- Functionally complete function
- Concept of Delay



Look ahead carry adder

Look Ahead Carry Adder

- **Ripple carry adder:**

$$T_d = (n-1)\,t_c + \max(t_c, t_s)$$

$T_d \rightarrow$ depends on n, higher the n higher will T_d.

- **Look Ahead Carry Adder.**

$A = a_3\ a_2\ a_1\ a_0$

$B = b_3\ b_2\ b_1\ b_0$

$+\quad c_3\ c_2\ c_1\ c_0$

$c_4\ S_3\ S_2\ S_1\ S_0$

$c_0 \Rightarrow$ known at $t = 0$

$$c_1 = a_0 b_0 + (a_0 \oplus b_0)\, c_0$$

$$c_2 = a_1 b_1 + (a_1 \oplus b_1)\, c_1$$

$$c_3 = a_2 b_2 + (a_2 \oplus b_2)\, c_2$$

$$c_4 = a_3 b_3 + (a_3 \oplus b_3)\, c_3$$

$$c_{i+1} = a_i b_i + (a_i \oplus b_i)\, c_i$$

$$x_i = a_i b_i$$

$$y_i = a_i \oplus b_i$$

$$c_{i+1} = x_i + y_i c_i$$

$$c_1 = x_0 + y_0 c_0$$

$$c_1 = a_0 b_0 + (a_0 \oplus b_0) c_0 = x_0 + y_0 c_0$$

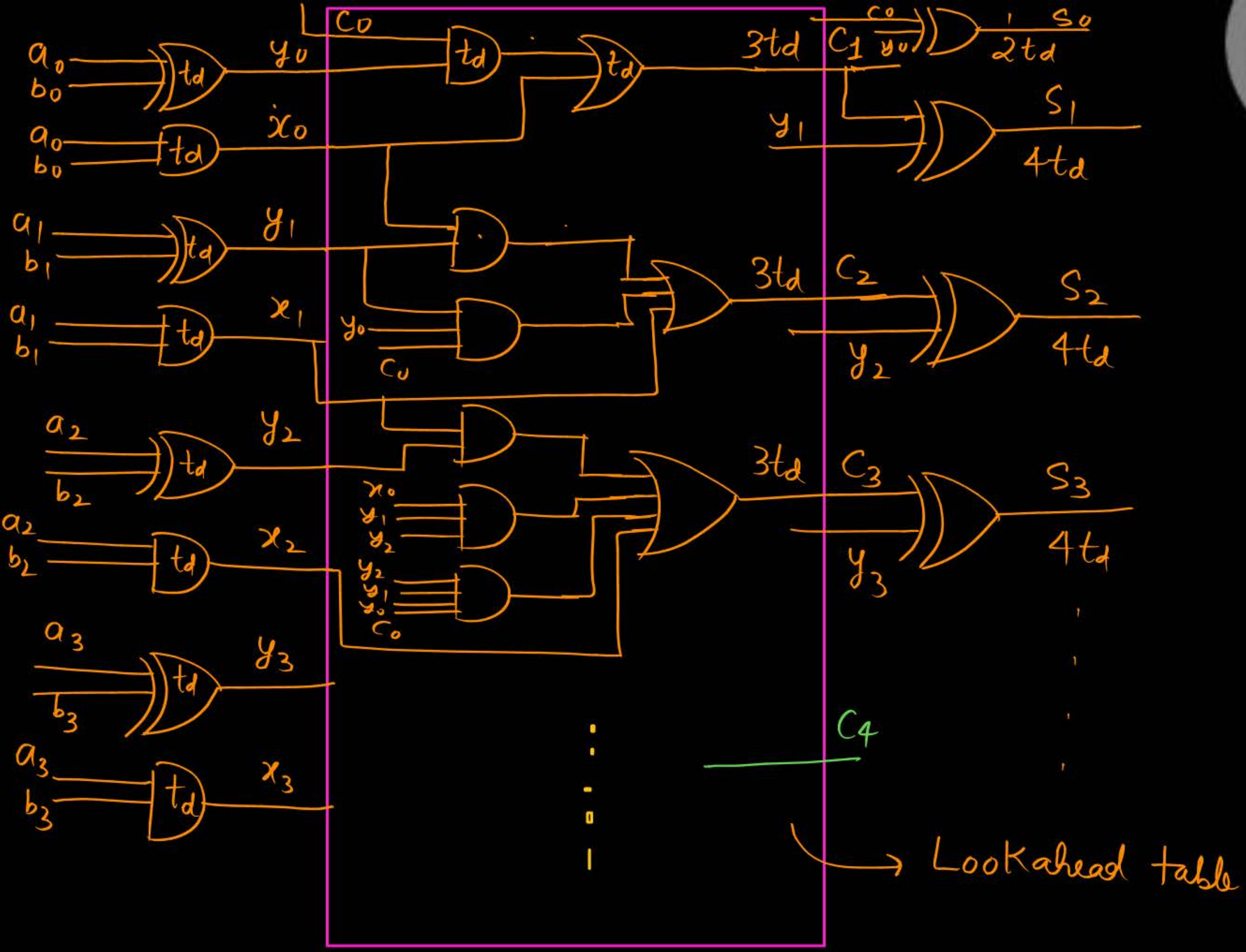
$$c_2 = x_1 + y_1 c_1 = x_1 + y_1 [x_0 + y_0 c_0]$$

$$c_2 = x_1 + x_0 y_1 + y_1 y_0 c_0$$

$$c_3 = x_2 + y_2 c_2 = x_2 + y_2 [x_1 + x_0 y_1 + y_1 y_0 c_0]$$

$$= x_2 + x_1 y_2 + x_0 y_1 y_2 + y_2 y_1 y_0 c_0$$

$$c_4 = x_3 + y_3 c_3 = x_3 + x_2 y_3 + x_1 y_2 y_3 + x_0 y_1 y_2 y_3 + y_3 y_2 y_1 y_0 c_0$$


a_0
b_0
td
y_0
c_0
td
td
3td
c_1
y_0
c_0
s_0
2td
a_0
b_0
td
x_0
y_1
s_1
4td
a_1
b_1
td
y_1
3td
c_2
s_2
4td
a_1
b_1
td
x_1
y_0
c_0
y_2
a_2
b_2
td
y_2
3td
c_3
s_3
4td
x_0
y_1
y_2
a_2
b_2
td
x_2
y_2
y_1
x_0
c_0
y_3
a_3
b_3
td
y_3
a_3
b_3
td
x_3
c_4
Lookahead table

- So by using Look ahead table carry o/ps can be generated parallely after 3 t_d time.
- All sum o/ps can be generated parallely in 4 t_d time.

Overall result of addition of two n-bit numbers can be generated in 4 t_d time.

$T_d = 4t_d$ independent of no. of bits

→ In look ahead table of n-bit addition →

no. of AND gates used are

$$N_D = 1 + 2 + 3 + \cdots + n$$

$$N_D = \frac{n(n+1)}{2}$$

→ In look ahead of n-bit addition

no. of OR gates used are

$$N_O = n$$

Q.a. A number is written in decimal no. system is: $(3\times512 + 2\times128 + 5\times8 + 7)_{10}$ then no. of 1's in its binary representation is: 8.

b. $(5\times2^{12} + 6\times2^{5} + 3\times2^{3} + 4)_{10}$ no. of 1's in binary representation is 7.

c. $(7\times2^{12} + 13\times2^{8} + 11)_{10}$, no. of 1's in its binary representation is 9.

d. A decimal no. is given as (101×2^{7}) then no. of 1's in its binary representation is _____.

a. $(3\times512+2\times128+5\times8+7)_{10}$

$=(3\times2^9+2\times2^7+5\times2^3+7)_{10}$

$=(3\times8^3+4\times8^2+5\times8^1+7\times8^0)_{10}$

$=(3457)_8$

$=(011\,100\,101\,111)_2$

$(3\times2^9+4\times2^6+7)_{10}$

$=3\times8^3+4\times8^2+7\times8^0$

$=(3407)_8$

b. $(5\times2^{12}+6\times2^{5}+3\times2^{3}+4)_{10}$

$(5\times8^{4}+3\times8^{2}+3\times8^{1}+4\times8^{0})_{10}$

$=(50334)_{8}$

$=(101\,000\,011\,011\,100)_{2}$

c. $(7\times2^{12}+13\times2^{8}+11)_{10}$

$=7\times16^{3}+13\times16^{2}+11\times16^{0}$

$=(7D0B)_{16}=(0111\,1101\,0000\,1011)_{2}$

- $(101\times2^{7})_{10}$

$=2^{7}[2^{6}+2^{5}+2^{2}+2^{0}]_{10}$

$=(2^{13}+2^{12}+2^{9}+2^{7})_{10}$

$[11001010000000]_{2}$ — 7 0's

$=2^{7}.[1100101]_{2}$

Conversion of one FF into another FF

available ff ⟶ desired ff

- Procedure :

→ Write down characteristic table of desired ff on i/p side

→ Excitation table of availabl ff on o/p side

→ Write down the excitation in SOP or POS format

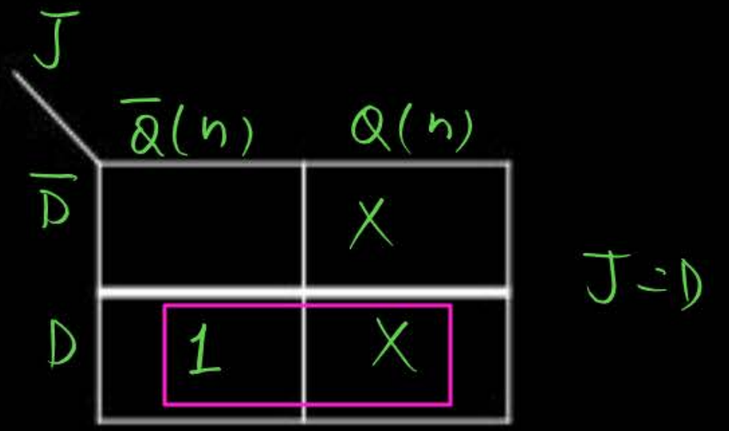
→ Solve it implement it

- JK to D Table :

D	Q(n)	Q(n+1)	J	K
0	0	0	0	X
0	1	0	X	1
1	0	1	1	X
1	1	1	X	0

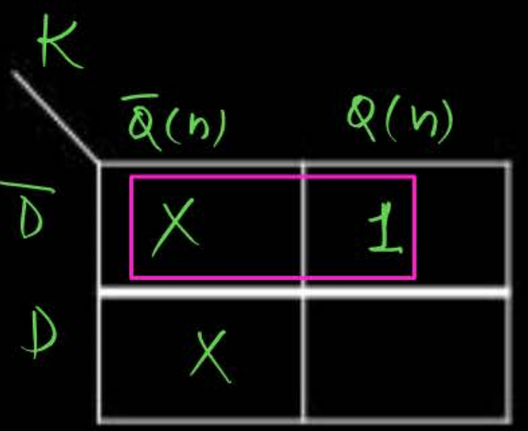
$J[D, Q(n)] = \Sigma(2) + d\Sigma(1,3)$

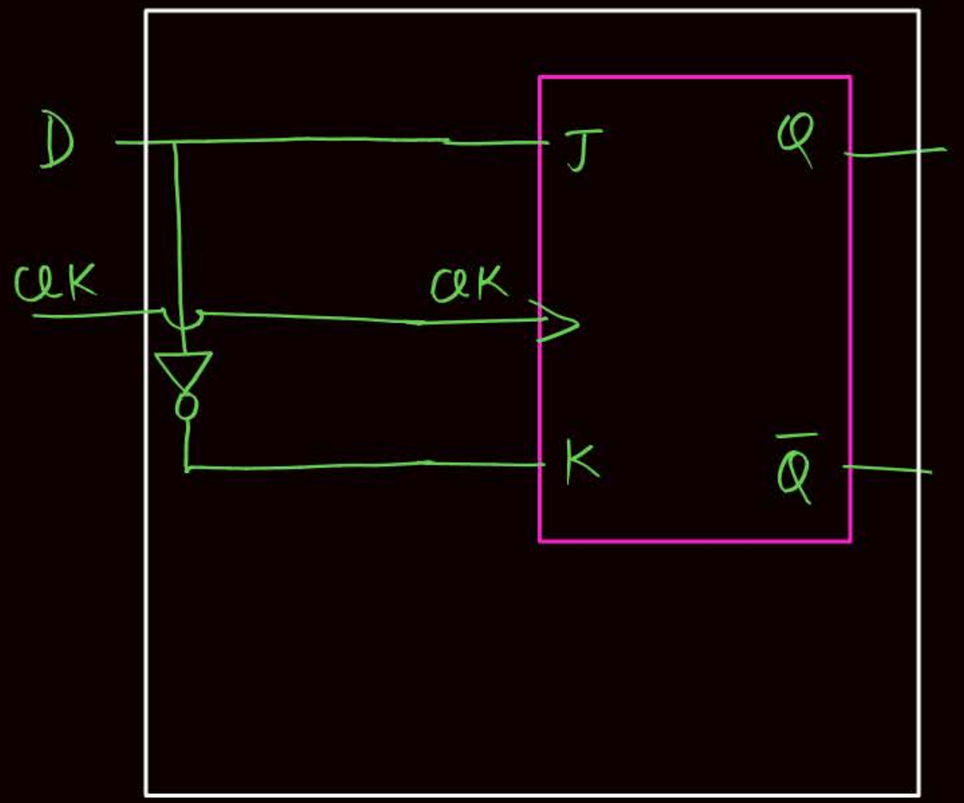
$K[D, Q(n)] = \Sigma(1) + d\Sigma(0,2)$



$J = D$

$K = \bar{D}$





$$Q(n+1) = J\bar{Q}(n) + \bar{K}Q(n)$$

$$= D\bar{Q}(n) + DQ(n)$$

$$Q(n+1) = D$$

- SR to JK :

Table :

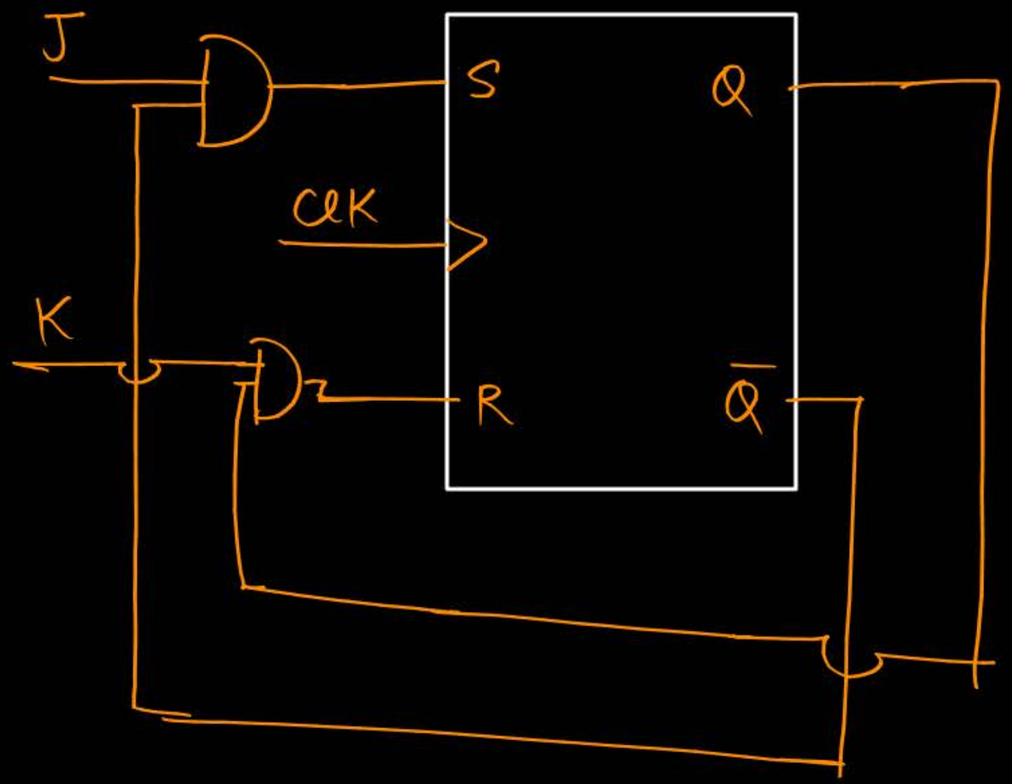
J	K	Q(n)	Q(n+1)	S	R
0	0	0	0	0	X
0	0	1	1	X	0
0	1	0	0	0	X
0	1	1	0	0	1
1	0	0	1	1	0
1	0	1	1	X	0
1	1	0	1	1	0
1	1	1	0	0	1

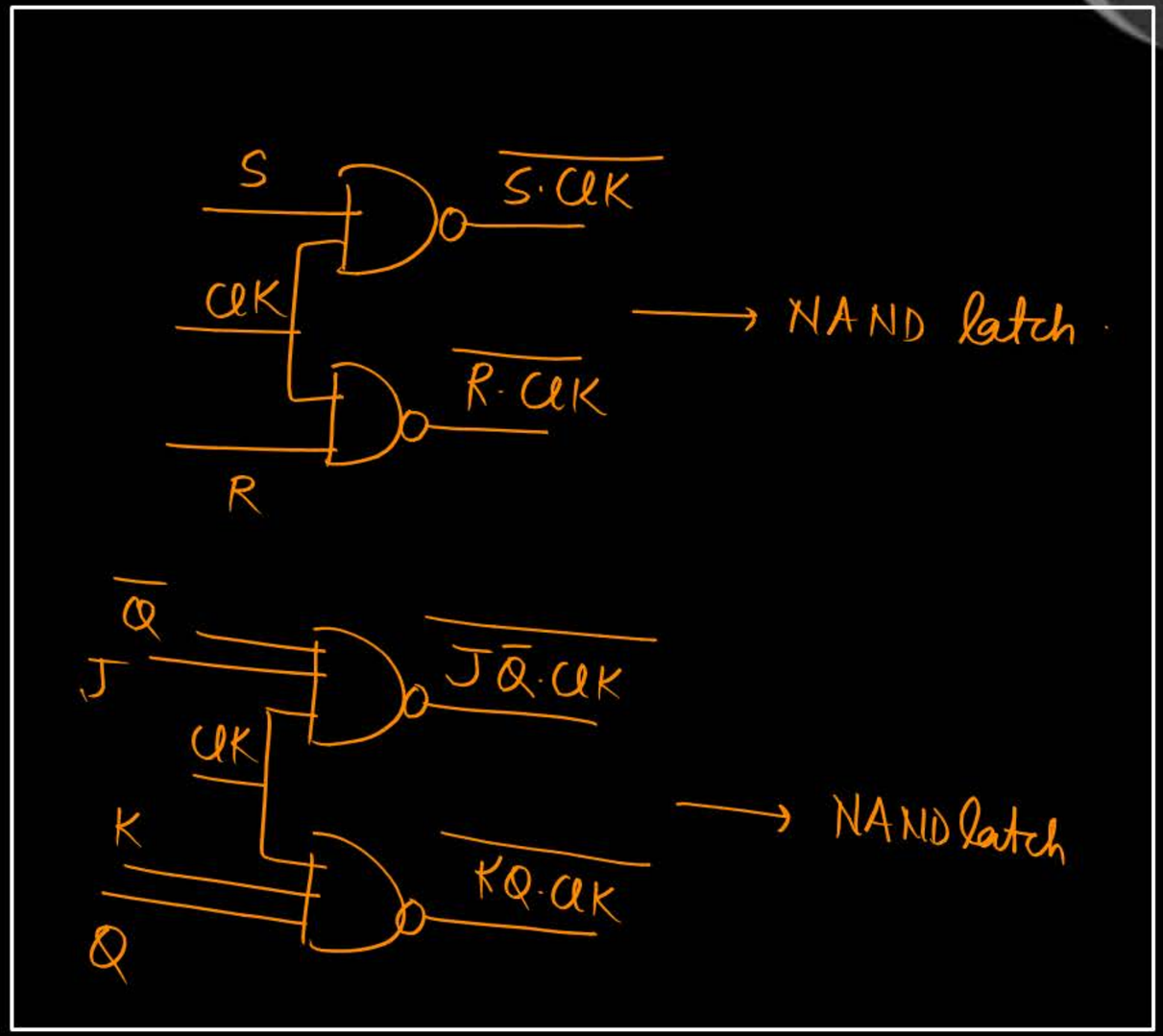


$S = J\bar{Q}(n)$



$R = KQ(n)$


J
S
Q
CLK
K
R
$\bar{Q}$


S
$\overline{S \cdot CLK}$
CLK
$\overline{R \cdot CLK}$
R
$\rightarrow$ NAND latch.
$\bar{Q}$
J
$\overline{J\bar{Q} \cdot CLK}$
CLK
K
$\overline{KQ \cdot CLK}$
Q
$\rightarrow$ NAND latch

All Conversion Results

SR - JK	$S = J\bar{Q}$, $R = KQ$
SR - D	$S = D$, $R = \bar{D}$
SR - T	$S = T\bar{Q}$, $R = TQ$
JK - SR	$J = S$, $K = R$
JK - D	$J = D$, $K = \bar{D}$
JK - T	$J = T$, $K = T$
D - JK	$D = J\bar{Q}(n) + \bar{K}Q(n)$
D - T	$D = T \oplus Q(n)$
T - JK	$T = J\bar{Q} + KQ$
T - SR	$T = S\bar{Q} + RQ$
T - D	$T = D \oplus Q(n)$



Topic : 2 Min Summary

- Look ahead carry adder
- Conversion of ff's.

Thank you

GW

Soldiers !

